

QUESTION 1

1.1) Here is the purpose of these Linux Directories:

- **/home:**
The purpose of /home is to contain the personal directory for all regular users. Every user gets their own subdirectory under /home. This is users store personal files, downloads and configured settings.
- **/etc:**
The purpose of /etc is to store system-wide configured files. These configured files have things like networking, users and applications located in there. An example would be /etc/passwd. This contains the user's account information.
- **/var:**
The purpose of /var is to contain variable data that frequently changes. This directory includes mail spools, log files, print queues and temporary files that are created by running services.
- **/bin:**
The purpose of /bin is to contain essential binary executables [programs]. These programs are command-line tools that are required for basic system operation such as ls, cp, mv and bash.
- **/usr:**
The purpose of /usr is to hold user related programs and data. It's one of the biggest directories that contain binaries like documentation, libraries and shared data for software installed in the system.

1.2)

/ETC DIRECTORY:

```
ning] - Oracle VirtualBox
View Input Devices Help

Jun 10 17:59
ubuntu@ubuntu: ~

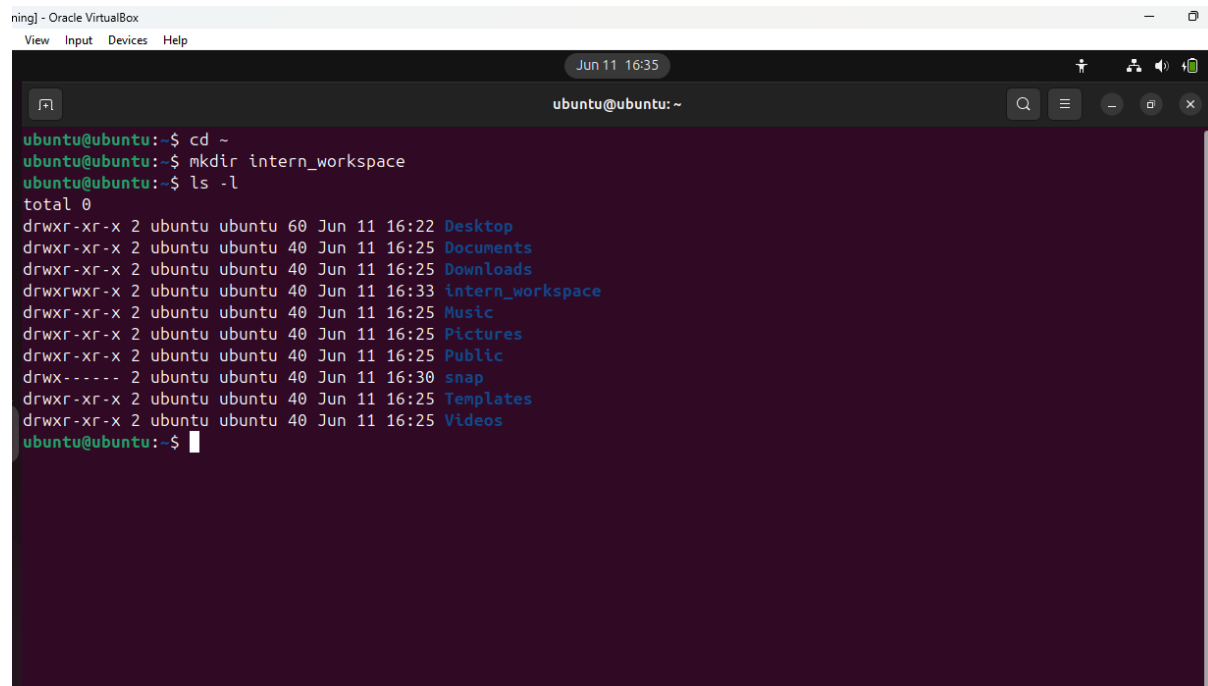
ubuntu@ubuntu:~$ ls /etc
ModemManager          gshadow               pm
NetworkManager        gshadow-              pnm2ppa.conf
PackageKit            gss                   polkit-1
UPower                gtk-2.0               ppp
X11                   gtk-3.0               profile
adduser.conf          hdparm.conf           profile.d
alsa                  host.conf             protocols
alternatives          hostid                pulse
anacrontab            hostname              python3
apg.conf              hosts                 python3.12
apm                   hosts.allow           rc0.d
apparmor              hosts.deny            rc1.d
apparmor.d            hp                    rc2.d
apport                ifplugd               rc3.d
apt                   init                  rc4.d
avahi                 init.d                rc5.d
bash.bashrc           initramfs-tools       rc6.d
bash_completion       inputrc               rcS.d
bindresvport.blacklist insserv.conf.d        request-key.conf
binfmt.d              ipp-usb               request-key.d
bluetooth             iproute2              resolv.conf
brlapi.key            issue                 rmt
brltty                issue.net             rpc
brltty.conf           kernel                rsyslog.conf
ca-certificates       kerneloops.conf       rsyslog.d
```

```
ning] - Oracle VirtualBox
View Input Devices Help

Jun 11 16:38
ubuntu@ubuntu: ~

dhcpd.conf             mke2fs.conf           sysctl.conf
dictionaries-common   ModemManager          sysctl.d
dpkg                   modprobe.d            sysstat
e2scrub.conf          modules               systemd
emacs                 modules-load.d        terminfo
environment            mtab                  thermald
environment.d          nanorc                timezone
ethertypes            netconfig             tmpfiles.d
fonts                 netplan               ubuntu-advantage
fprintd.conf          network               ucf.conf
fstab                 networkd-dispatcher   udev
fuse.conf              NetworkManager        udisks2
fwupd                 networks              ufw
gai.conf              newt                  update-manager
gdb                   nftables.conf         update-motd.d
gdm3                  nsswitch.conf         update-notifier
geoclue               openvpn               UPower
ghostscript           opt                   usb_modeswitch.conf
glvnd                  os-release            usb_modeswitch.d
gnome                 PackageKit            vconsole.conf
gnome-remote-desktop  pam.conf              vim
gnutls                pam.d                 vtrgb
groff                 papersize             vulkan
group                 passwd                wgetrc
group-                passwd-               wpa_supplicant
grub.d                pcmcia                X11
```

INTERN_WORKSPACE:



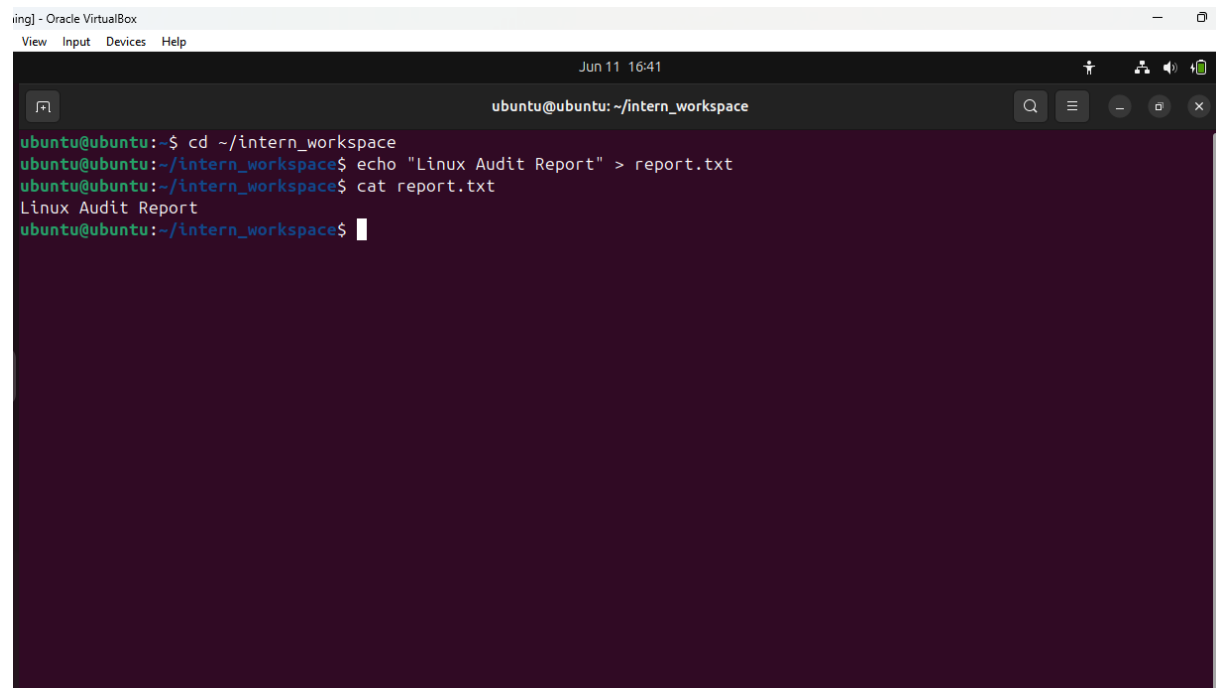
The screenshot shows a terminal window titled 'ningj - Oracle VirtualBox' with a menu bar (View, Input, Devices, Help) and a status bar (Jun 11 16:35). The terminal prompt is 'ubuntu@ubuntu: ~'. The user enters the following commands:

```
ubuntu@ubuntu:~$ cd ~
ubuntu@ubuntu:~$ mkdir intern_workspace
ubuntu@ubuntu:~$ ls -l
```

The output of the 'ls -l' command is as follows:

```
total 0
drwxr-xr-x 2 ubuntu ubuntu 60 Jun 11 16:22 Desktop
drwxr-xr-x 2 ubuntu ubuntu 40 Jun 11 16:25 Documents
drwxr-xr-x 2 ubuntu ubuntu 40 Jun 11 16:25 Downloads
drwxrwxr-x 2 ubuntu ubuntu 40 Jun 11 16:33 intern_workspace
drwxr-xr-x 2 ubuntu ubuntu 40 Jun 11 16:25 Music
drwxr-xr-x 2 ubuntu ubuntu 40 Jun 11 16:25 Pictures
drwxr-xr-x 2 ubuntu ubuntu 40 Jun 11 16:25 Public
drwx----- 2 ubuntu ubuntu 40 Jun 11 16:30 snap
drwxr-xr-x 2 ubuntu ubuntu 40 Jun 11 16:25 Templates
drwxr-xr-x 2 ubuntu ubuntu 40 Jun 11 16:25 Videos
```

The terminal prompt is now 'ubuntu@ubuntu:~\$'.



The screenshot shows a terminal window titled 'ningj - Oracle VirtualBox' with a menu bar (View, Input, Devices, Help) and a status bar (Jun 11 16:41). The terminal prompt is 'ubuntu@ubuntu: ~/intern_workspace'. The user enters the following commands:

```
ubuntu@ubuntu:~/intern_workspace$ cd ~/intern_workspace
ubuntu@ubuntu:~/intern_workspace$ echo "Linux Audit Report" > report.txt
ubuntu@ubuntu:~/intern_workspace$ cat report.txt
```

The output of the 'cat report.txt' command is:

```
Linux Audit Report
```

The terminal prompt is now 'ubuntu@ubuntu:~/intern_workspace\$'.

```
ning] - Oracle VirtualBox
View Input Devices Help

Jun 11 16:51

ubuntu@ubuntu: ~/intern_workspace

ubuntu@ubuntu:~$ cd ~/intern_workspace
ubuntu@ubuntu:~/intern_workspace$ echo "Linux Audit Report" > report.txt
ubuntu@ubuntu:~/intern_workspace$ cat report.txt
Linux Audit Report
ubuntu@ubuntu:~/intern_workspace$ mv report.txt /tmp/
ubuntu@ubuntu:~/intern_workspace$ ls /tmp/
gdm3-config-err-8PHvdb
report.txt
snap-private-tmp
systemd-private-cd460b4e32f74517a767b7bab115f5b1-colord.service-g8teJy
systemd-private-cd460b4e32f74517a767b7bab115f5b1-ModemManager.service-W0Mcsb
systemd-private-cd460b4e32f74517a767b7bab115f5b1-polkit.service-5f5ceM
systemd-private-cd460b4e32f74517a767b7bab115f5b1-power-profiles-daemon.service-idXdys
systemd-private-cd460b4e32f74517a767b7bab115f5b1-switcheroo-control.service-ziVt0T
systemd-private-cd460b4e32f74517a767b7bab115f5b1-systemd-logind.service-5lGUpD
systemd-private-cd460b4e32f74517a767b7bab115f5b1-systemd-oond.service-tCSMrI
systemd-private-cd460b4e32f74517a767b7bab115f5b1-systemd-resolved.service-ltV83F
systemd-private-cd460b4e32f74517a767b7bab115f5b1-systemd-timesyncd.service-WtFFxi
systemd-private-cd460b4e32f74517a767b7bab115f5b1-upower.service-AazSmR
ubuntu@ubuntu:~/intern_workspace$
```

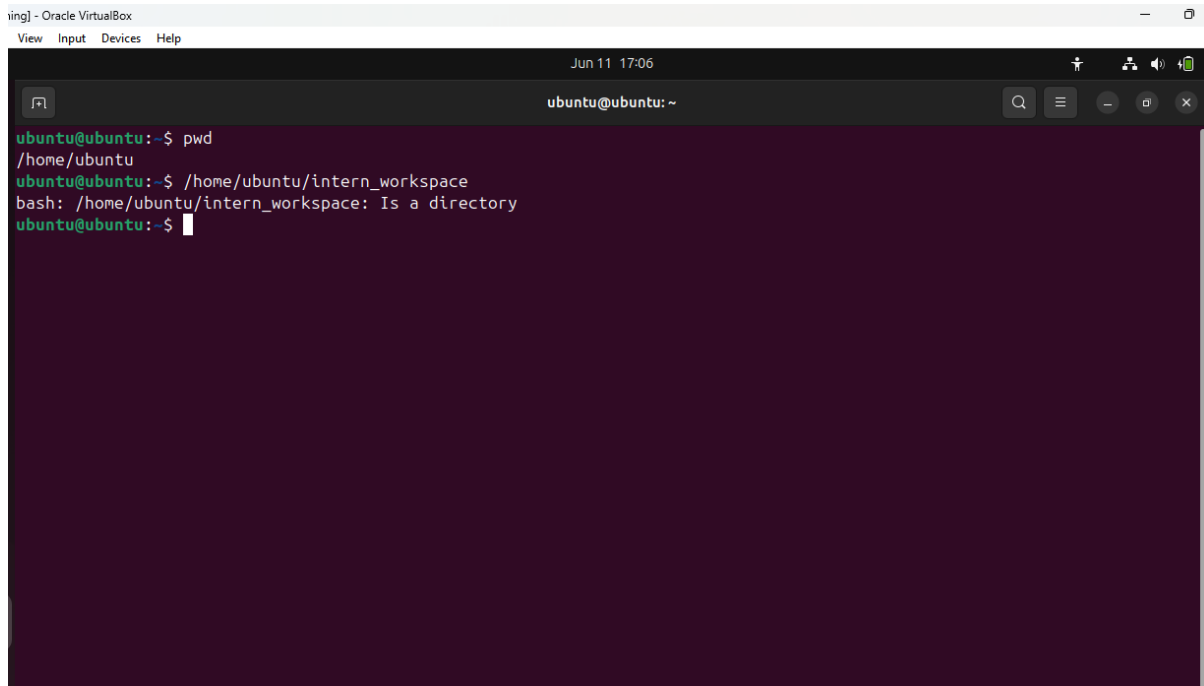
```
ning] - Oracle VirtualBox
View Input Devices Help

Jun 11 16:59

ubuntu@ubuntu: ~/intern_workspace

ubuntu@ubuntu:~$ cd ~/intern_workspace
ubuntu@ubuntu:~/intern_workspace$ echo "Linux Audit Report" > report.txt
ubuntu@ubuntu:~/intern_workspace$ cat report.txt
Linux Audit Report
ubuntu@ubuntu:~/intern_workspace$ mv report.txt /tmp/
ubuntu@ubuntu:~/intern_workspace$ ls /tmp/
gdm3-config-err-8PHvdb
report.txt
snap-private-tmp
systemd-private-cd460b4e32f74517a767b7bab115f5b1-colord.service-g8teJy
systemd-private-cd460b4e32f74517a767b7bab115f5b1-ModemManager.service-W0Mcsb
systemd-private-cd460b4e32f74517a767b7bab115f5b1-polkit.service-5f5ceM
systemd-private-cd460b4e32f74517a767b7bab115f5b1-power-profiles-daemon.service-idXdys
systemd-private-cd460b4e32f74517a767b7bab115f5b1-switcheroo-control.service-ziVt0T
systemd-private-cd460b4e32f74517a767b7bab115f5b1-systemd-logind.service-5lGUpD
systemd-private-cd460b4e32f74517a767b7bab115f5b1-systemd-oond.service-tCSMrI
systemd-private-cd460b4e32f74517a767b7bab115f5b1-systemd-resolved.service-ltV83F
systemd-private-cd460b4e32f74517a767b7bab115f5b1-systemd-timesyncd.service-WtFFxi
systemd-private-cd460b4e32f74517a767b7bab115f5b1-upower.service-AazSmR
ubuntu@ubuntu:~/intern_workspace$ rm /tmp/report.txt
ubuntu@ubuntu:~/intern_workspace$ ls /tmp/
gdm3-config-err-8PHvdb
snap-private-tmp
systemd-private-cd460b4e32f74517a767b7bab115f5b1-colord.service-g8teJy
systemd-private-cd460b4e32f74517a767b7bab115f5b1-ModemManager.service-W0Mcsb
systemd-private-cd460b4e32f74517a767b7bab115f5b1-polkit.service-5f5ceM
```

```
ubuntu@ubuntu:~/intern_workspace$ rm /tmp/report.txt
ubuntu@ubuntu:~/intern_workspace$ ls /tmp/
gdm3-config-err-8PHvdb
snap-private-tmp
systemd-private-cd460b4e32f74517a767b7bab115f5b1-colord.service-g8teJy
systemd-private-cd460b4e32f74517a767b7bab115f5b1-ModemManager.service-W0Mcsb
systemd-private-cd460b4e32f74517a767b7bab115f5b1-polkit.service-5f5ceM
systemd-private-cd460b4e32f74517a767b7bab115f5b1-power-profiles-daemon.service-idXdys
systemd-private-cd460b4e32f74517a767b7bab115f5b1-switcheroo-control.service-ziVt0T
```



```
ing] - Oracle VM VirtualBox
View Input Devices Help
Jun 11 17:06
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ pwd
/home/ubuntu
ubuntu@ubuntu:~$ /home/ubuntu/intern_workspace
bash: /home/ubuntu/intern_workspace: Is a directory
ubuntu@ubuntu:~$
```

QUESTION 2

2.1) A shell script is a text file that contains a series of commands written for a Unix/Linux shell. It is executed by the shell interpreter and automates repetitive tasks that would otherwise be typed manually in the command line.

A shell script can automate tasks like this:

- **TASK AUTOMATION:**
Shell scripts automate routine system tasks like file backups, user management, log cleanups and software installations. An example would be a backup script that can compress files, move them to a directory, add timestamps and upload them to the cloud without user intervention.
- **SCHEDULING WITH CRON:**
Scripts can be scheduled using cron which is a time-based job that schedules in Unix/Linux. This will ensure that the script will run at regular intervals when set. For example, 14:00pm daily.

- **EFFICIENCY:**

Shell scripts run faster and more efficiently than manual input. This saves time and reduces workload especially for tasks like daily server backups such as in TechNet Solutions case.

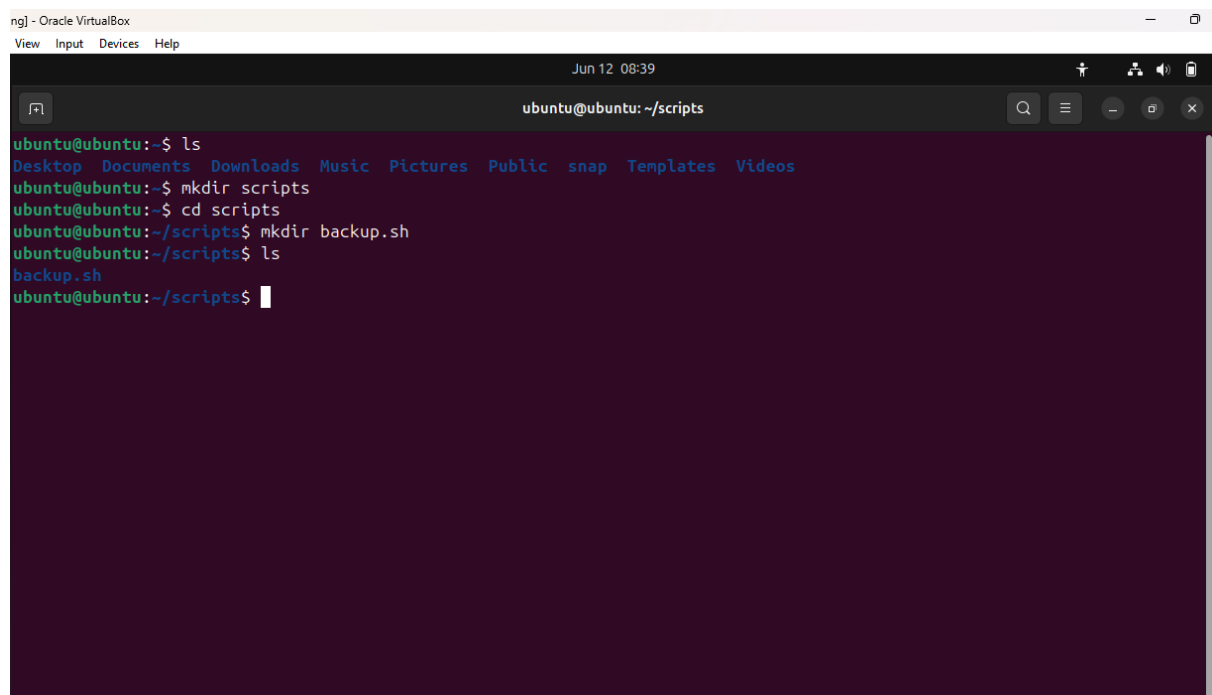
- **ERROR HANDLING:**

Shell scripts can include logging and error checking. This ensures that if there's any problems like missing files or upload failures that are detected and recorded it will get sorted out.

- **CONSISTENCY AND RELIABILITY:**

By automating the steps with a script, you end up eliminating any sort of human error. This ensures that tasks are completed consistently and accurately each time.

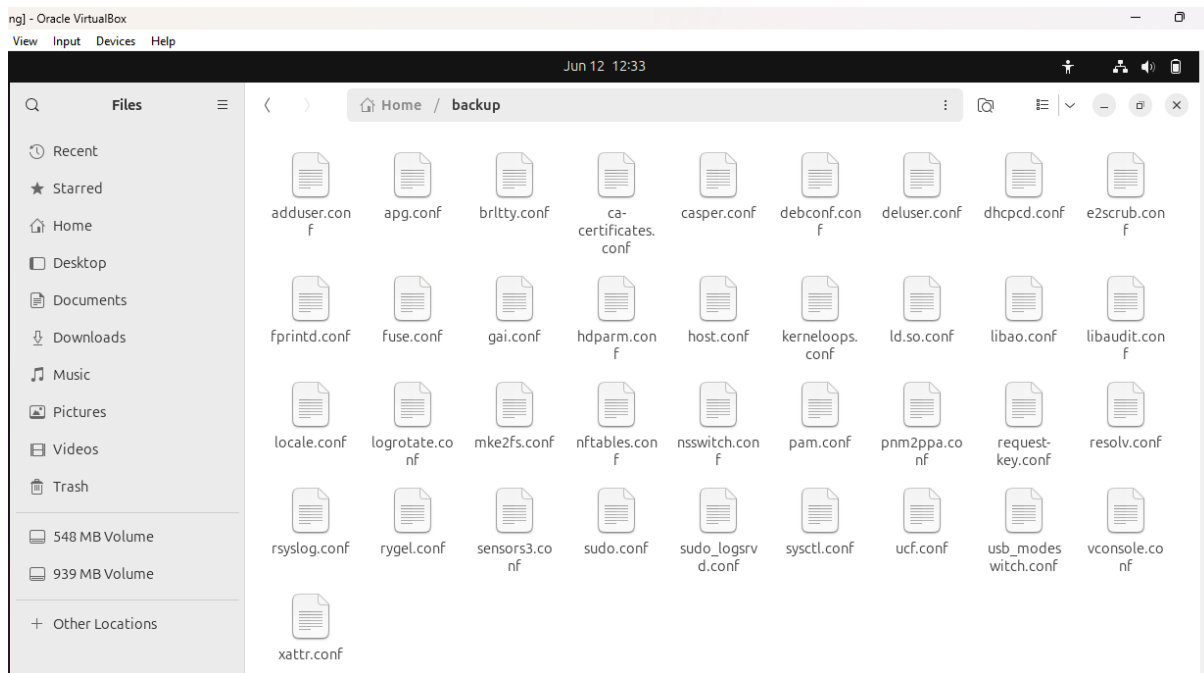
2.2)



```
ng] - Oracle VM VirtualBox
View Input Devices Help
Jun 12 08:39
ubuntu@ubuntu: ~/scripts
ubuntu@ubuntu:~$ ls
Desktop Documents Downloads Music Pictures Public snap Templates Videos
ubuntu@ubuntu:~$ mkdir scripts
ubuntu@ubuntu:~$ cd scripts
ubuntu@ubuntu:~/scripts$ mkdir backup.sh
ubuntu@ubuntu:~/scripts$ ls
backup.sh
ubuntu@ubuntu:~/scripts$
```

```
ingl - Oracle VirtualBox
View Input Devices Help
Jun 12 12:21
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ mkdir -p ~/backup
ubuntu@ubuntu:~$ cp /etc/*.conf ~/backup/
ubuntu@ubuntu:~$ echo "Backup completed successfully!"
Backup completed successfully!
ubuntu@ubuntu:~$
```

```
ingl - Oracle VirtualBox
View Input Devices Help
Jun 12 12:26
ubuntu@ubuntu: ~/scripts
ubuntu@ubuntu:~$ mkdir -p ~/backup
ubuntu@ubuntu:~$ cp /etc/*.conf ~/backup/
ubuntu@ubuntu:~$ echo "Backup completed successfully!"
Backup completed successfully!
ubuntu@ubuntu:~$ cd scripts
ubuntu@ubuntu:~/scripts$ chmod +x backup.sh
ubuntu@ubuntu:~/scripts$ ./backup.sh
bash: ./backup.sh: Is a directory
ubuntu@ubuntu:~/scripts$
```



2.3) The key advantages of using shell scripting in System Administration are:

- **TASK AUTOMATION:**

With shell scripting it automates repetitive administrative tasks such as reducing manual effort and human error. An example would be a system administrator writing a shell script to:

- Backup configuration files daily
- Update software packages automatically
- Clean temporary files weekly.

- **EFFICIENT SYSTEM MANAGEMENT**

With shell scripting it can perform complex tasks quickly by combining multiple commands and making the system management more efficient. An example would be where a script can:

- Monitor disk usage
- Log the event in a monitoring file
- Monitor disk usage

QUESTION 3

3.1)

```
iningj - Oracle VirtualBox
View Input Devices Help
Jun 12 18:20
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ sudo adduser name_surname
info: Adding user 'name_surname' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'name_surname' (1002) ...
info: Adding new user 'name_surname' (1002) with group 'name_surname (1002)' ...
info: Creating home directory '/home/name_surname' ...
info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for name_surname
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user 'name_surname' to supplemental / extra groups 'users' ...
info: Adding user 'name_surname' to group 'users' ...
ubuntu@ubuntu:~$
```

```
ning] - Oracle VirtualBox
View Input Devices Help

Jun 12 18:24

ubuntu@ubuntu: ~

ubuntu@ubuntu:~$ sudo adduser name_surname
fatal: The user 'name_surname' already exists.
ubuntu@ubuntu:~$ sudo groupadd staff
groupadd: group 'staff' already exists
ubuntu@ubuntu:~$ sudo usermod -aG staff name_surname
ubuntu@ubuntu:~$ sudo touch /home/name_surname/welcome.txt
ubuntu@ubuntu:~$ sudo chown name_surname:staff /home/name_surname/welcome.txt
ubuntu@ubuntu:~$
```

```
ning] - Oracle VirtualBox
View Input Devices Help

Jun 12 18:27

ubuntu@ubuntu: ~

ubuntu@ubuntu:~$ sudo adduser name_surname
fatal: The user 'name_surname' already exists.
ubuntu@ubuntu:~$ sudo groupadd staff
groupadd: group 'staff' already exists
ubuntu@ubuntu:~$ sudo usermod -aG staff name_surname
ubuntu@ubuntu:~$ sudo touch /home/name_surname/welcome.txt
ubuntu@ubuntu:~$ sudo chown name_surname:staff /home/name_surname/welcome.txt
ubuntu@ubuntu:~$ sudo chmod 600 /home/name_surname/welcome.txt
ubuntu@ubuntu:~$ id name_surname
uid=1002(name_surname) gid=1002(name_surname) groups=1002(name_surname),50(staff),100(users)
ubuntu@ubuntu:~$ ls -l /home/name_surname/welcome.txt
ls: cannot access '/home/name_surname/welcome.txt': Permission denied
ubuntu@ubuntu:~$
```

3.2)

- **CHMOD:** The purpose of chmod command is to set or change the read, write and execute permissions for files and directories. An example would be **chmod 600 welcome.txt**. This sets permissions so that only the owner can read and write.

- **CHOWN:**

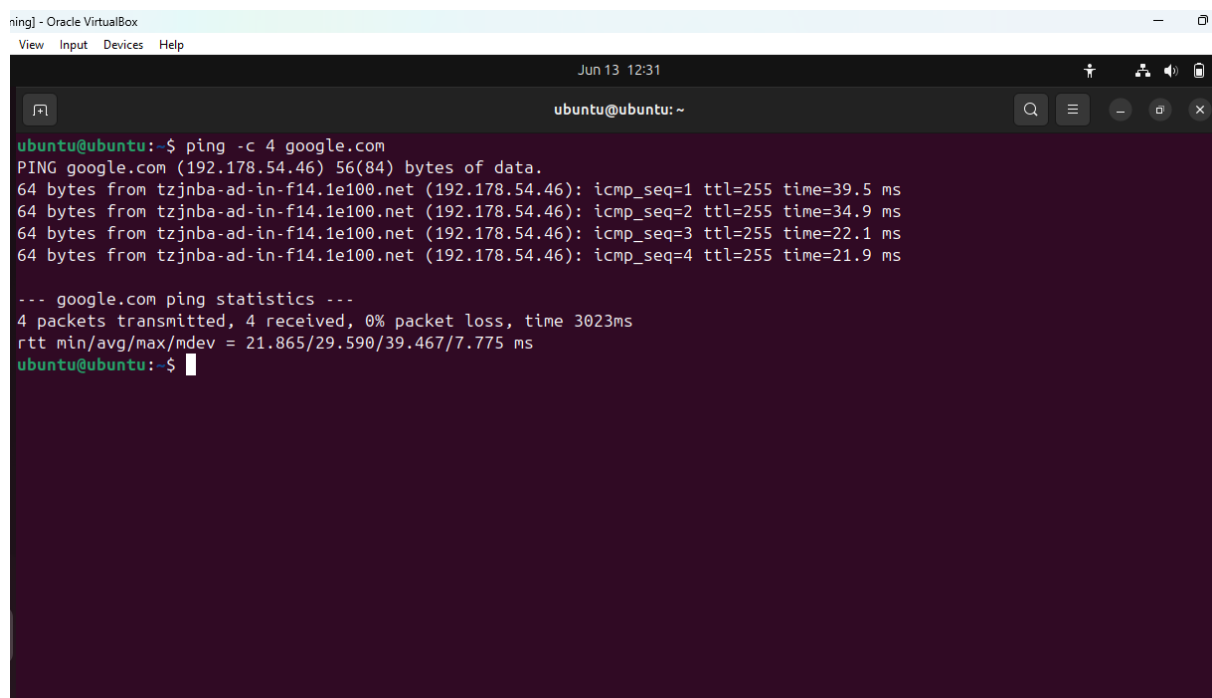
The purpose of chown command is to change the owner or group of a file or directory. An example would be `chown john_doe:staff welcome.txt`. This makes john_doe the owner and staff the group owner of the file.

- **CHGRP:**

The purpose of the chgrp command is to specifically change only the group ownership of a file or directory. An example would be `chgrp staff welcome.txt`. This sets the group of the file to staff without changing the owner's name.

QUESTION 4

4.1a)



```
ning] - Oracle VirtualBox
View Input Devices Help
Jun 13 12:31
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ ping -c 4 google.com
PING google.com (192.178.54.46) 56(84) bytes of data:
64 bytes from tzjnba-ad-in-f14.1e100.net (192.178.54.46): icmp_seq=1 ttl=255 time=39.5 ms
64 bytes from tzjnba-ad-in-f14.1e100.net (192.178.54.46): icmp_seq=2 ttl=255 time=34.9 ms
64 bytes from tzjnba-ad-in-f14.1e100.net (192.178.54.46): icmp_seq=3 ttl=255 time=22.1 ms
64 bytes from tzjnba-ad-in-f14.1e100.net (192.178.54.46): icmp_seq=4 ttl=255 time=21.9 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3023ms
rtt min/avg/max/mdev = 21.865/29.590/39.467/7.775 ms
ubuntu@ubuntu:~$
```

- Ping -c 4 google.com sends 4 ICMP echo requests to Google.com to check internet connectivity.
- Packet loss or high latency of >100ms indicates a network issue.
- If it fails to resolve the domain, there may be DNS issues.

b)

```
ningj - Oracle VirtualBox
View Input Devices Help
64 bytes from tzjnba-ad-in-f14.1e100.net (192.178.54.46): icmp_seq=2 ttl=255 time=34.9 ms
64 bytes from tzjnba-ad-in-f14.1e100.net (192.178.54.46): icmp_seq=3 ttl=255 time=22.1 ms
64 bytes from tzjnba-ad-in-f14.1e100.net (192.178.54.46): icmp_seq=4 ttl=255 time=21.9 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3023ms
rtt min/avg/max/mdev = 21.865/29.590/39.467/7.775 ms
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:96:90:ee brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86050sec preferred_lft 86050sec
    inet6 fd17:625c:f037:2:1dff:a21a:348a:8585/64 scope global temporary dynamic
        valid_lft 86131sec preferred_lft 14131sec
    inet6 fd17:625c:f037:2:a00:27ff:fe96:90ee/64 scope global dynamic mngtmpaddr
        valid_lft 86131sec preferred_lft 14131sec
    inet6 fe80::a00:27ff:fe96:90ee/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$
```

- Ip a list all network interfaces and their IP addresses.
- The interface must be UP with a valid IP address.
- We end up looking for things like inet address. Example: 192.168.1.100.

c)

```
ningj - Oracle VirtualBox
View Input Devices Help
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:96:90:ee brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86050sec preferred_lft 86050sec
    inet6 fd17:625c:f037:2:1dff:a21a:348a:8585/64 scope global temporary dynamic
        valid_lft 86131sec preferred_lft 14131sec
    inet6 fd17:625c:f037:2:a00:27ff:fe96:90ee/64 scope global dynamic mngtmpaddr
        valid_lft 86131sec preferred_lft 14131sec
    inet6 fe80::a00:27ff:fe96:90ee/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$ sudo netstat -tuln
sudo: netstat: command not found
ubuntu@ubuntu:~$ sudo ss -tuln
Netid      State      Recv-Q     Send-Q     Local Address:Port      Peer Address:Port      Process
udp        UNCONN    0           0           0.0.0.0:60580            0.0.0.0:*
udp        UNCONN    0           0           0.0.0.0:5353             0.0.0.0:*
udp        UNCONN    0           0           127.0.0.54:53            0.0.0.0:*
udp        UNCONN    0           0           127.0.0.53%lo:53         0.0.0.0:*
udp        UNCONN    0           0           [::]:5353                [::]:*
udp        UNCONN    0           0           [::]:49164                [::]:*
tcp        LISTEN    0           4096        127.0.0.1:631            0.0.0.0:*
tcp        LISTEN    0           4096        127.0.0.53%lo:53         0.0.0.0:*
tcp        LISTEN    0           4096        127.0.0.54:53            0.0.0.0:*
tcp        LISTEN    0           4096        [::]:631                  [::]:*
```

- `sudo netstat -tuln` / `sudo ss -tuln` shows the services currently listening and also shows active connections.
- It also checks for any unexpected open ports that may indicate malicious services or misconfiguration.

d)

```

ingj - Oracle VM VirtualBox
View Input Devices Help

Jun 13 12:39
ubuntu@ubuntu: ~

ubuntu@ubuntu:~$ traceroute google.com
traceroute to google.com (192.178.54.46), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  170.287 ms  1.369 ms  1.268 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
  
```

- A traceroute displays each hop a packet takes to reach a destination.
- It displays `***` in the middle to indicate network or ISP routing issues.

e)

```
mingj - Oracle VirtualBox
View Input Devices Help
24 * * *
25 * * *
26 * * *
27 * * *
28 * * *
29 * * *
30 * * *
ubuntu@ubuntu:~$ dig google.com

; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 57478
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                86      IN      A      192.178.54.46

;; Query time: 179 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Jun 13 12:40:41 UTC 2025
;; MSG SIZE rcvd: 55

ubuntu@ubuntu:~$
```

- Dig google.com confirms that the DNS is working

4.2)

```
mingj - Oracle VirtualBox
e View Input Devices Help
Jun 13 16:19
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            298M  1.8M  296M   1% /run
/dev/sr0         6.0G   6.0G   0 100% /cdrom
/cow             1.5G   55M   1.4G   4% /
tmpfs            1.5G   8.0K   1.5G   1% /dev/shm
tmpfs            5.0M   8.0K   5.0M   1% /run/lock
tmpfs            1.5G    0     1.5G   0% /tmp
tmpfs            298M  152K  298M   1% /run/user/1000

ubuntu@ubuntu:~$
```

- By using df, it displays the disk usage.
- By using -h, provides human readability.
- With /dev/sr0, it usually is a root partition.
- If Use% is above 80%, it means the disk has to be cleaned up.

```

iningl - Oracle VirtualBox
View Input Devices Help

Jun 13 16:21
ubuntu@ubuntu: ~

ubuntu@ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           298M  1.8M  296M   1% /run
/dev/sr0        6.0G  6.0G   0 100% /cdrom
/cow            1.5G   55M   1.4G   4% /
tmpfs           1.5G  8.0K  1.5G   1% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
tmpfs           1.5G   0 1.5G   0% /tmp
tmpfs           298M  152K  298M   1% /run/user/1000

ubuntu@ubuntu:~$ ps aux --sort=-%mem | head -n 6
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
ubuntu    3140  8.4  11.3 3361672 343980 ?        Ssl   16:14   0:33 /usr/bin/gnome-shell
ubuntu    3931  0.4   3.5 912120 107768 ?        Ssl   16:15   0:01 /usr/libexec/xdg-desktop-portal-gnome
ubuntu    3435  0.4   3.5 989344 107740 ?        Sl    16:14   0:01 /usr/libexec/evolution-data-server/evolution-alarm-no
tify
ubuntu    2460  2.5   3.3 357956 103240 tty2    Sl+   16:13   0:10 /usr/lib/xorg/Xorg vt2 -displayfd 3 -auth /run/user/1
000/gdm/Xauthority -nolisten tcp -background none -noreset -keeptty -novtswitch -verbose 3
ubuntu    3289  0.4   3.2 814956 99056 ?        Sl    16:14   0:01 /usr/libexec/mutter-x11-frames
ubuntu@ubuntu:~$

```

- This shows all processes sorted by memory usage
- Head -n 6 indicates that you must include the header and top 5 processes
- This helps monitor apps like chrome or large builds for example for optimizations or restart.

```
ning] - Oracle VirtualBox
View Input Devices Help

Jun 13 17:29

ubuntu@ubuntu: ~

ubuntu@ubuntu:~$ sudo apt update && sudo apt install python3 -y
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble Release
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [868 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [291 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [160 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/main Icons (48x48) [13.4 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main Icons (64x64) [20.0 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [7068 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe i386 Packages [527 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [849 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [187 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1118 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [17.0 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1138 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [473 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/restricted i386 Packages [17.6 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [236 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [238 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [161 kB]
```

```
ning] - Oracle VirtualBox
View Input Devices Help

Jun 13 17:29

ubuntu@ubuntu: ~

Get:32 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:33 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [592 B]
Get:34 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [39.2 kB]
Get:35 http://archive.ubuntu.com/ubuntu noble-backports/main i386 Packages [31.8 kB]
Get:36 http://archive.ubuntu.com/ubuntu noble-backports/main Translation-en [8676 B]
Get:37 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7076 B]
Get:38 http://archive.ubuntu.com/ubuntu noble-backports/main Icons (48x48) [9521 B]
Get:39 http://archive.ubuntu.com/ubuntu noble-backports/main Icons (64x64) [11.2 kB]
Get:40 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [272 B]
Get:41 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [27.1 kB]
Get:42 http://archive.ubuntu.com/ubuntu noble-backports/universe i386 Packages [14.1 kB]
Get:43 http://archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [16.5 kB]
Get:44 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [16.4 kB]
Get:45 http://archive.ubuntu.com/ubuntu noble-backports/universe Icons (48x48) [20.4 kB]
Get:46 http://archive.ubuntu.com/ubuntu noble-backports/universe Icons (64x64) [28.7 kB]
Get:47 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1304 B]
Get:48 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:49 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 11.6 MB in 60s (194 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
289 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2).
python3 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 289 not upgraded.
```

- This command installs python3 and its dependencies.
- -y confirms the installation

```
ning] - Oracle VirtualBox
View Input Devices Help
Get:65 http://archive.ubuntu.com/ubuntu noble-backports/universe Icons (48x48) [20.4 kB]
Get:66 http://archive.ubuntu.com/ubuntu noble-backports/universe Icons (64x64) [28.7 kB]
Get:67 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1304 B]
Get:68 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:69 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 11.6 MB in 60s (194 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
289 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2).
python3 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 289 not upgraded.
ubuntu@ubuntu:~$ sudo apt update && sudo remove python3 -y
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble Release
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
289 packages can be upgraded. Run 'apt list --upgradable' to see them.
sudo: remove: command not found
ubuntu@ubuntu:~$
```

- This command was supposed to remove python3 but failed to do so.

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